

CLINTON, MO, USA

WMO: 720636

Lat: 38.350N

Lon: 93.683W

Elev: 251

StdP: 98.34

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00222

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.0	-12.1	-19.9	0.7	-13.8	-17.2	0.9	-10.4	10.8	1.4	9.6	1.6	1.9	10	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	36.1	24.7	33.8	25.1	32.6	25.0	27.2	32.2	26.4	31.5	25.6	30.8	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	22.1	30.5	25.0	20.7	29.7	24.0	19.4	28.5	87.6	31.5	83.8	31.3	80.1	31.2	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	8.0	7.2	-18.7	37.9	4.9	3.1	-22.2	40.1	-25.1	41.9	-27.9	43.7	-31.4	45.9
			-19.2	28.4	4.7	1.1	-22.6	29.1	-25.3	29.8	-28.0	30.4	-31.4	31.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	0.1	1.4	8.5	13.8	18.9	24.5	26.1	24.7	21.0	14.1	7.7	2.6
	DBStd	10.42	6.41	6.40	6.41	4.97	4.82	3.22	3.35	3.24	4.19	5.38	5.66	5.46
	HDD10.0	1047	312	247	104	20	2	0	0	0	0	21	104	236
	HDD18.3	2538	567	474	312	148	54	1	0	1	19	154	319	488
	CDD10.0	2392	4	7	57	135	279	436	498	456	331	147	37	6
	CDD18.3	840	0	0	6	13	73	187	239	198	101	22	1	0
	CDH23.3	8470	0	0	37	126	609	1934	2705	1970	899	183	6	0
	CDH26.7	3553	0	0	3	21	166	809	1294	873	344	42	1	0
Wind	WSAvg	2.9	3.2	3.4	3.7	3.7	2.9	2.8	2.1	1.9	2.1	2.7	3.1	3.1
Precipitation	PrecAvg	1109	45	53	73	124	139	139	109	90	111	87	81	58
	PrecMax	1487	153	142	158	401	335	262	353	187	330	233	274	180
	PrecMin	734	7	4	23	40	36	43	15	19	34	12	18	12
	PrecStd	199	31	35	38	65	67	62	79	43	72	56	63	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	21.3	27.2	28.9	32.2	36.0	39.0	37.8	34.9	30.0	24.0	20.2
		MCWB	13.4	13.1	18.8	20.1	23.7	25.6	24.4	23.7	23.9	23.2	16.4	15.9
	2%	DB	16.0	17.6	23.8	27.1	30.0	33.6	37.1	35.0	32.4	27.7	21.3	17.1
		MCWB	10.6	12.7	16.7	18.2	22.4	25.5	24.4	24.8	24.4	19.6	15.4	13.1
	5%	DB	12.5	14.1	22.0	24.1	28.6	32.5	34.1	32.9	30.9	25.2	18.8	13.7
		MCWB	8.0	9.4	15.0	16.9	22.2	25.2	25.4	24.6	23.7	19.2	14.1	9.9
	10%	DB	8.9	11.2	18.7	22.5	27.3	31.3	32.7	31.4	28.0	22.8	17.1	11.4
		MCWB	5.9	7.0	13.3	16.2	21.1	24.6	25.5	24.2	22.4	17.4	12.4	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	16.3	19.9	21.6	25.0	27.7	28.4	28.4	25.8	23.5	18.5	17.0
		MCDB	17.2	19.7	25.6	27.2	30.4	32.5	33.2	34.0	31.3	28.4	22.0	19.3
	2%	WB	11.1	13.8	18.0	19.6	23.7	26.7	27.3	26.9	25.1	22.0	16.6	13.5
		MCDB	14.7	16.6	22.8	24.4	28.3	32.0	32.2	32.0	30.8	25.6	19.2	15.8
	5%	WB	8.4	9.9	16.0	18.4	22.6	25.8	26.5	25.8	24.2	20.3	14.9	10.9
		MCDB	11.6	15.0	20.5	22.9	27.1	31.2	31.6	30.6	29.3	23.3	18.2	13.0
	10%	WB	5.9	7.1	13.8	17.2	21.7	25.1	25.7	25.1	23.3	18.7	12.8	8.2
		MCDB	9.3	10.6	16.8	20.9	26.1	30.4	30.9	29.8	27.3	22.3	16.3	10.4
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.0	11.1	11.7	10.7	10.5	10.7	11.1	11.6	12.3	11.3	9.4
		MCDBR	14.4	14.8	14.5	14.1	12.0	11.2	12.3	13.1	12.5	13.7	14.2	13.6
		MCWBR	9.8	10.6	8.3	7.9	6.3	5.4	4.3	5.3	5.6	7.2	9.6	9.6
	5% WB	MCDBR	13.3	14.1	12.3	11.5	10.8	10.5	10.1	10.7	11.3	10.7	12.3	12.0
		MCWBR	9.9	11.1	7.8	7.6	6.1	5.6	4.9	5.4	5.7	6.5	9.6	10.1
Clear-Sky Solar Irradiance	taub	0.300	0.313	0.339	0.387	0.428	0.431	0.446	0.449	0.389	0.346	0.309	0.294	
	taud	2.459	2.429	2.410	2.290	2.238	2.264	2.246	2.236	2.364	2.469	2.515	2.503	
	Ebn at Noon	883	916	922	891	856	851	834	823	857	867	867	867	
	Edh at Noon	81	95	107	129	139	135	137	135	112	91	76	73	
All-Sky Solar Radiation	RadAvg	2.24	2.93	3.85	4.93	5.67	6.46	6.42	5.68	4.83	3.47	2.40	1.91	
	RadStd	0.25	0.36	0.33	0.31	0.50	0.52	0.40	0.34	0.32	0.37	0.26	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page